

Decision Making and Steps in Business Analytics

Decision Making in Business Analytics

Decision making in business analytics is the process of selecting the best possible action by collecting, analyzing, and interpreting data in a business environment.

Instead of guessing, businesses use:

- Reports
- Dashboards
- Forecasts
- Statistical models
- Optimization techniques

The main aim is to convert:

Data → Information → Insight → Action

Decision making is important because it helps:

- Improve business performance
- Reduce risks
- Increase profits
- Improve customer satisfaction

Business analytics provides accurate and data-driven decisions instead of assumptions.

Types of Decisions in Business

Business analytics supports different levels of decision making.

1. Strategic Decisions

Strategic decisions are long-term and high-level decisions taken by top management.

These decisions affect the future direction of the company.

Examples

- Entering a new market
- Launching a new product
- Acquiring another company

Example:

- Tesla is expanding electric vehicle production globally.
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2. Tactical Decisions

Tactical decisions are medium-term decisions related to departmental planning and resource management.

Examples

- Marketing budget planning
- Employee scheduling
- Sales target setting

3. Operational Decisions

Operational decisions are short-term routine decisions related to daily business activities.

Examples

- Day-to-day operations
 - Repeated regularly
 - Staff work schedules
 - Daily production planning
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Steps in Business Analytics for Decision Making

Business analytics follows a structured process to convert raw data into useful business decisions.

1. Define the Business Problem

The first step is identifying and clearly defining the business problem.

The organization must understand:

- What is the problem?
 - What decision should be made?
 - What is the business goal?
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2. Understand the Business Context

Analysts study the business environment and background of the problem.

Factors Studied

- Competitors
 - Customer expectations
 - Government regulations
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3. Identify Data Requirements

The required data for solving the problem is identified.

Data Requirements Include

- Sales data
 - Customer information
 - Cost details
 - Demand data
 - Time period
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4. Collect the Data

Data is gathered from different internal and external sources.

Sources of Data

Internal Sources	External Sources
• Sales databases • Finance records • CRM systems • ERP systems	• Surveys • Government reports • Social media • Websites

Types of Data

Structured Data	Unstructured Data
• Tables • Spreadsheets	• Emails • Images & Videos

5. Clean and Prepare the Data

Raw data is cleaned and organized before analysis.

Activities

- Removing duplicate data
 - Handling missing values
 - Correcting errors
 - Standardizing formats
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6. Explore the Data

Data exploration helps analysts understand patterns and trends.

Techniques Used

- Charts and graphs
 - Trend analysis
 - Reports & Dashboard
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7. Apply Analytical Models

Different analytical techniques are used to generate insights and predictions.

Type of Analytics	Main Question	Example
Descriptive Analytics	What happened?	Sales reports
Diagnostic Analytics	Why did it happen?	Root cause analysis
Predictive Analytics	What may happen?	Sales forecasting
Prescriptive Analytics	What should be done?	Optimization techniques

8. Make the Decision

Managers select the best solution based on analytical results.

Factors Considered

- Business objectives
- Cost
- Risk
- Feasibility
- Profitability

Benefits of Decision Making in Business Analytics

1. Better business decisions
2. Increased profits
3. Faster problem solving
4. Better customer satisfaction

Applications

- Banking sector
- Healthcare industry
- Retail business
- Manufacturing companies
- E-commerce industries

Example:

- Amazon uses business analytics for customer recommendations and delivery optimization.

OLAP (Online Analytical Processing)

OLAP (Online Analytical Processing) is a technology used in business analytics to analyze large amounts of data from different perspectives quickly and efficiently.

OLAP allows users to:

- View data in different dimensions
- Generate reports
- Analyze trends

OLAP is mainly used in:

- Business Intelligence (BI)
- Data Warehousing
- Decision Support Systems (DSS)

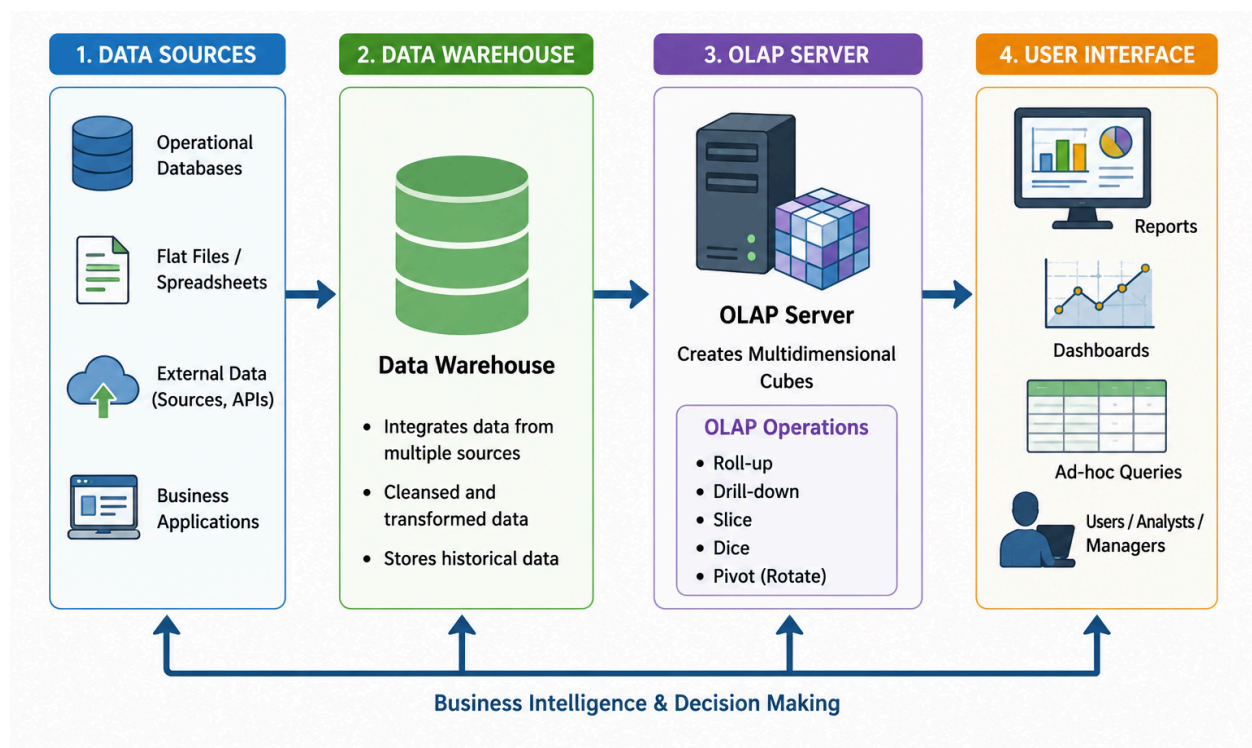
Features of OLAP

1. Multidimensional Analysis
2. Fast Data Processing
3. Complex Calculations
4. Data Aggregation
5. Interactive Analysis

OLAP Architecture

OLAP mainly works with:

1. Data Sources
2. Data Warehouse
3. OLAP Server
4. User Interface



Working of OLAP

Step 1: Data Collection - Data is collected from various sources.

Step 2: Data Storage - Data is stored in a data warehouse.

Step 3: OLAP Processing - The OLAP server processes multidimensional data.

Step 4: Data Analysis - Users generate reports, charts, and dashboards for analysis.

OLAP Operations

1. Roll-Up - It summarizes detailed data into higher-level information.

Example: Daily sales → Monthly sales.

2. Drill-Down - It moves from summarized data to detailed data.

Example: Yearly sales → Monthly sales → Daily sales.

3. Slice - Slice selects a single dimension from the data cube.

Example: Viewing sales data for only one year.

4. Dice - Dice selects data from multiple dimensions.

Example: Sales data for a specific product and region.

5. Pivot (Rotate) - Pivot changes the view of data for better analysis.

Example: Changing rows into columns in reports.

Applications of OLAP

1. Sales analysis
2. Financial reporting
3. Budget planning
4. Marketing analysis
5. Customer analysis
6. Supply chain management

Example:

- Amazon uses OLAP for customer and sales analysis.